



The efficient and stable planning for interrupted supply chain with dual-sourcing strategy: a robust optimization approach considering decision maker's risk attitude[☆]

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ABSTRACT

In the face of the complex market environment and the in-depth development of globalization, ensuring the supply chain's stability and cost leadership by developing a differentiated dual-source procurement strategy is extremely important. The present study contemplates the supply chain as the research object. It also considers disruption uncertainty and decision-makers' risk attitudes. To minimize costs and shortages, we build a dual-objective robust optimization model for a dual-sourcing strategy under interruption risk. The authors aim to explore efficient and robust planning under supply disruption uncertainty in dual-sourcing strategies by different risk attitudes. The research employs the augmented ε -constraint method to solve the proposed model by transforming the model into a single-objective mixed-integer programming model. Additionally, our study uses enterprise examples to prove the feasibility and effectiveness of the model. Moreover, it conducts risk attitude analysis, dual objective function trade-off analysis, and superiority analysis of procurement strategies. Finally, we put forward some management insights to provide a new decision-making idea for real supply chain enterprises in the case of supply interruption risk.

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1. Introduction

The supply chain enterprise's uncertain internal factors, unpredictable external events, and unpredictable business environment lead to supply interruption, bringing economic losses and damaging the enterprise's reputation. Dual-sourcing is a practical and cost-effective approach to mitigate interruption risk in the supply chain by reducing shortages of minimal economic cost [1], which refers to selecting two suppliers in supply chain procurement to share the interruption risk [2]. Decision-makers need to carry out relatively effective and stable order planning, which is the key to implementing a dual-sourcing strategy under the risk of supply interruption. In addition, different risk attitudes of decision-makers may affect ordering plans. Therefore, this study explores efficient and robust ordering planning in dual-sourcing strategies by decision-makers with different risk attitudes under supply disruption uncertainty.

Mitigation strategies for supply interruption must establish flexibility to adapt to supply chain changes; one approach is dual-sourcing [2–7]. A well-known example is Nokia and Ericsson upstream supply factory outage incident [8]. After this disruption, Nokia designed a quick fix, using a backup supplier to produce the chips in just five days. Meanwhile, Ericsson eliminated backup suppliers to save costs before this disruption, so it endured a huge loss and withdrew from the mobile phone production market. Further, this example also tells us how to purchase from the primary and backup suppliers, which must be considered in dual-sourcing research.

Decision-makers' risk attitude significantly impacts dual-source subscription planning [4,9,10]. A risk-neutral decision-maker will ultimately adopt the dual-sourcing strategy [10]. As the decision-maker becomes risk-averse, the allocation of orders to the leading supplier will be reduced [4]. Risk attitudes are variable and occur on a continuum ranging from risk avoidance through risk neutrality to risk preference. However, because the research above focuses on using dual-sourcing tactics from a single risk attitude, it is crucial to examine how changes in decision makers' risk attitudes affect their purchase decisions.

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Additionally, implementing economically and effectively a dual-sourcing strategy for supply outages is the research direction of many scholars [1,5,11–14]. Silbermayr and Minner [13] are concerned about all costs in the supply chain to adopt the dual-source ordering strategy effectively. Also, some studies on dual-sourcing strategy use game theory [14] or a real-option method [1] to investigate how to effectively dual-source ordering at the risk of disruption. However, suppose the dual-sourcing strategy is only economically and effectively adopted. In that case, it is often easy to ignore the difficulties in implementing the strategy caused by the interruption, such as uncertainty of interruption and shortage of demand the market will endure. As a result, we must consider how to efficiently balance costs and the scarcity of downstream demand in the supply chain caused by disruption, as well as how to successfully implement a dual-sourcing strategy in the face of various disruption concerns.

Our study examines how to perform an efficient and stable dual-source subscription program with different decision-makers risk attitudes under supply disruption uncertainty. First, we construct a dual-objective robust optimization model for dual-sourcing planning and express the uncertain interruption probability as an interval [3]. The conservatism parameter in a robust model, in a sense, indicates decision-makers' attitudes about risk. Second, we apply and analyze the model using the real enterprise data. We confirm the viability of the dual-sourcing strategy, examine the suggested model's efficacy in cost and market shortfall balancing, and investigate the model's stability under uncertainty and the impact of the decision maker's risk attitude on the strategy. Third, we discuss the feasibility and applicability of the proposed model and summarize the full text and prospects. The model efficiently conducts dual-sourcing planning under disruption uncertainty and different risk attitudes, which is relatively novel in dual-sourcing research.

The rest of the paper is developed as follows. Section 2 briefly reviews the relevant literature. Section 3 starts with the problem statement and constructs a dual-objective robust optimization model for dual-sourcing strategy under the supply interruption risk and uses the augmented ϵ -constraint method to deal with this model. Section 4 carries on the numerical example analysis and obtains the practical and managerial insights. Further, Section 5 discusses the feasibility and practicability of the model proposed. Finally, Section 6 concludes the paper.

2. Literature Review

In the case of supply disruptions, dual-sourcing is a complex problem [3], which is solved by considering different aspects with various objectives. This section reviews some research literature on dual-sourcing around supplier reliability, outage scenario settings, risk attitudes, a comparative study on objectives and robust methods, and additional supplements on dual-sourcing with an uncertainty of supply disruptions.

In the dual-sourcing proposed by Gupta et al. [15], the expected total market output decreases with supplier reliability, affecting supplier choice. Manufacturers pick the lower-cost supplier over the higher-cost supplier in the case of two suppliers with equivalent reliability, but for two suppliers with varying reliability is the opposite [14]. According to specific research, such as [16], the buyer can affect supply reliability by offering incentives to lower the risk of supply interruption, like promoting the contract's endogenous and exogenous activities of dual procurement. Supplier reliability is often correlated with the probability of supply disruptions. Zhu and Fu [60] proposed that an unreliable supplier tends to have a greater probability of disruption.

Shu et al. [11] showed that in a dual-sourcing model with service constraints, the retailer's optimal decision is influenced by

the joint outage probability of primary and standby suppliers. Xue et al. [6] used the all-or-nothing type of disruption; the supplier cannot provide any order once an interruption occurs. Meanwhile, Silbermayr and Minner [13] assumed that interrupts have on/off cycles and that the cycles are exponentially distributed. Simultaneously, much literature has set outage probability parameters. Schmitt and Snyder [17] set the probability of interruption and demonstrated that a medium probability of interruption causes the single-period model to underestimate the impact of interruption risk, resulting in a significant increase in cost. Similarly, Ledari et al. [18] considered two suppliers with a certain probability of interruption. In contrast, Chen et al. [19] conjectured that interruption has recoverability. They defined "interruption" as the failure of suppliers to deliver orders fully, including non-delivery, which will initially lead to a massive loss of production capacity; however, the level of production capacity will eventually improve.

Li and Li [9] found that in the risk-neutral case, the total order quantity has no relation to the procurement cost and the reliability of reliable and unreliable suppliers, respectively. The risk attitude in economics and finance is quoted here through VaR and CVaR modeling or utility function modeling [10]. Using the expected utility function, Giri [4] studied the ordering strategy of risk-averse retailers and found that an increase in risk aversion reduces the optimal order quantity but increases the safety inventory. Meanwhile, Sawik [20] took conditional risk value at risk measures and compared risk-neutral and risk-averse solutions that optimize supply chains' average and worst-case performance. Furthermore, risk-averse decision-makers tend to adopt a more "conservative" policy and obtain a larger order quantity by setting appropriate service level restrictions [60].

Robust optimization is an emerging concept gradually developed in recent decades; it is a kind of uncertain optimization method closely related to the robust control concepts [21–25]. Scenario-based and interval data robust optimization methods are the most widely used [26–30]. In particular, the interval robust optimization method can reflect decision makers' risk attitude in managing uncertainty when it is not highly dependent on the occurrence probability of nominal scenarios [31–34]. Jabbarzadeh et al. [31] expressed a time-dependent conservative value in the interval data robustness model, based on which the decision maker's risk attitude can be reflected. In the robust model established by José Alem and Morabito [32], the risk value assumed is determined by the decision-makers' risk preference. Zhu et al. [33] established numerous intervals based on uncertain objectives, and the decision-makers' risk preferences determined the order of these intervals.

Additionally, robust optimization is an excellent method for studying the influence of supply-side/demand-side uncertainty on the supply chain [35–41]. Jabbarzadeh et al. [36] proposed a hybrid robust stochastic optimization model for designing a supply chain resilient to supply, demand, and facility interruptions. Later, Jabbarzadeh et al. [35] established a robust model for common supply interruptions and demand changes. In addition, Mirzapour Al-e-hashem et al. [42] proposed a robust multi-objective mixed-integer nonlinear programming model to deal with demand and cost uncertainty. Meanwhile, Lu et al. [37] solved the problem of location design of reliable facilities by applying distributionally robust optimization. The purpose is to minimize the expected cost with a given marginal probability of outages in the worst-case distribution while allowing the outage associated with uncertain joint distribution. The robust models developed by Goli et al. [43], Mirzapour Al-e-hashem et al. [42], and Alinezhad et al. [44] are customer-focused and dual-objective planning cost-strategies in which maximizing customer satisfaction involves minimizing demand shortage. While the dual-objective robust approach put forth by Tirko-

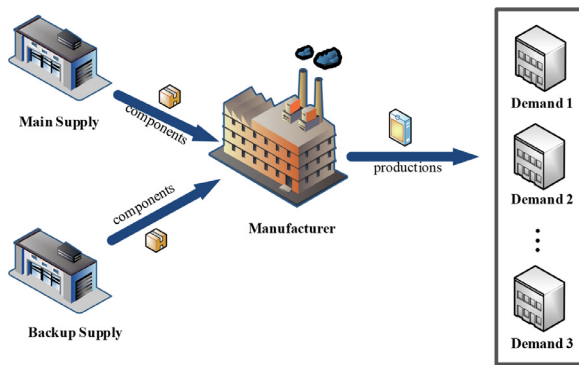


Fig. 1. Schematic structure of the proposed dual-sourcing in the supply chain.

laee et al. [45] aims to minimize cost and maximize supply reliability.

An augmented ε -constraint method combined with a robust optimization model can transform dual-objective problems with non-linear problems into linear models, which are easy to solve [26,31]. Jabbarzadeh et al. [31] applied the augmented ε -constraint method to study the impact of decision makers' risk preferences and delay strategies on economic and environmental goals under uncertain demand. A dual-objective robust optimization model proposed by Sun et al. [26] discusses the location–traffic–allocation emergency logistics problem under the uncertainties of casualties, the number of relief supplies, and transportation time.

In Table 1, we summarize the characteristics of procurement literature under disruption and related models, and our paper can be distinguished from previous work in the following ways. First, in contrast to other research, we set supplier reliability to the degree of change in the probability of interruption: the more significant the change, the less reliable the supplier. So reliability and outage probability fuse to better represent supply uncertainty and study stable dual-sourcing in uncertainty sets. Furthermore, unlike earlier studies on single risk attitudes, we considered the impact of various modifications in decision-makers' risk attitudes on dual-sourcing strategies. We also perfectly captured the risk attitude using the robust optimization method. Lastly, we suggested a reliable procedure for creating a dual-objective reliable dual-sourcing model that reduces costs and shortages. Our approach is pretty innovative in the area of dual-sourcing studies.

The main contributions of this paper are as follows. Initially, a dual-objective dual-sourcing model based on a robust optimization method is established to minimize costs and shortages. This method yields optimal order allocation results that satisfy various uncertainty sets. In addition, the robust method identifies different risk attitudes, making it simple to research how diverse risk attitudes affect the outcomes of order allocation. Ultimately, our proposed model is implemented in a real study case, which is applicable and feasible.

3. Model formulation and solution

3.1. Problem formulation

3.1.1. Problem definition

Purchasing is an essential process for the supply side and is related to the demand side [48]. Dual-sourcing is considered in the supply chain (Fig. 1): a manufacturing supply chain has one manufacturer, multiple demand markets, and two suppliers, one of which is a relatively cheap but independent main supply, and the other is an expensive but dependent backup supply. Considering the process of purchasing, production, and inventory, our goal is to minimize the total cost and market shortage quantity to solve

Table 1
Characteristics of procurement literature under disruption and related models.

References	Elements of dual-sourcing research		Decision makers' attitudes to risk	Interrupt setting	Objective/s	Method
	Reliability					
Gupta et al. [15]	Exogenous		-	All-or-nothing	Profit	Game theory
Zeng and He [14]	Related to outage probability		-	Outage probability	Profit	Game theory
Zhu et al. [46]	Exogenous		Risk neutral	Outage probability	Profit	Profit decision model
Shu et al. [11]	Exogenous		-	Outage probability	Profit	Profit decision model
Schmitt and Snyder [17]	Exogenous		-	All-or-nothing	Cost	News vendor model
Ledari et al. [18]	Exogenous		-	Outage probability	Profit	News vendor policy model
Giri [4]	Exogenous		Risk aversion	Quantified by an exponential utility function	Cost	Cost decision model
Mirzapour Al-e-hashem et al. [42]	-		-	-	Cost; Shortage quantity	Robust method
Sawik [20]	Exogenous		Risk neutral, risk-averse	Outage probability	Cost; Customer service Level	Stochastic Mixed integer programming approach
Zhu et al. [33]	Exogenous		-	Outage probability	Cost	Cost decision model
Li and Li [9]	Related to outage probability		Risk aversion	Outage probability	Cost	Cost decision model
He et al. [1]	Exogenous		-	Outage probability	Profit	Real option method
Han et al. [47]	Exogenous		-	Outage probability	Profit	Real option method
The present paper	Degree of change for outage probability		Various changes in risk attitude	Outage probability	Cost; Shortage quantity	Interval robust optimization method

order quantity from two suppliers, respectively. Meanwhile, we set the following assumptions.

- (1) Only one product is selected, one component in the manufacturer's procurement and product; other products and components meet the demand.
- (2) All products can be delivered to the demand market.
- (3) The loss rate of the two suppliers during regular operation is zero.
- (4) The purchasing period is the same as the production cycle: sourcing at the beginning of the period and producing and delivering before the end of the period.
- (5) The initial inventory level of components and products is zero in the first cycle.
- (6) The backup supply is more dependent but more expensive than the main supply. The former has less production constraint and interruption probability than the latter, but the latter has higher procurement and fixed order costs.

3.1.2. Notation description

The following notations (Table 2) are defined for formulating the proposed model.

3.1.3. Formulation of the normal scenario model

$$\min z_1 = \sum_t \sum_i \lambda_{it} g_{it} + \sum_t \sum_i c_{it} x_{it} + R(Pb_t) + R(Cb_t) + \sum_t PM_{jt} e_t \quad (1)$$

$$\min z_2 = \sum_t SP_{jt} \quad (2)$$

subject to:

$$Cb_{t+1} = \sum_i x_{it} p_{it} + Cb_t - kPM_{jt}, \quad \forall t \in T-1, j \in J, \quad (3)$$

$$Pb_t = Cb_t + \sum_i x_{it} p_{it} - \sum_j PM_{jt} - \sum_j D_{jt}, \quad \forall t \in T, \quad (4)$$

$$\sum_t \sum_i \lambda_{it} x_{it} = \sum_t \sum_j D_{jt}, \quad (5)$$

$$\sum_j PM_{jt} = V_t \cdot \gamma_t, \quad \forall t \in T, \quad (6)$$

$$\sum_t (Q_{it} - x_{it}) \geq 0, \quad \forall i \in \{1, 2\}, \quad (7)$$

$$\sum_i x_{it} + Cb_t \leq CH_t, \quad \forall t \in T, \quad (8)$$

$$\sum_j PM_{jt} + Pb_t \leq PH_t, \quad \forall t \in T, \quad (9)$$

$$SP_{j(t+1)} = \begin{cases} -Pb_t, & Pb_t < 0, \\ 0, & \text{otherwise,} \end{cases} \quad \forall t \in T, j \in J, \quad (10)$$

$$x_{it} \geq 0 \text{ and integer}, \quad \forall t \in T, \quad (11)$$

$$p_{it} \in [0, 1], \quad \forall t \in T, \quad (12)$$

$$R(Pb_t) = \max \left(Pb_t \cdot Ph_t, \sum_j \sum_i p_{it} a_{jt} SP_{jt} \right), \quad \forall t \in T, \quad (13)$$

$$R(Cb_t) = \max \left(\sum_t Cb_t \cdot Ch_t, \sum_j \sum_i p_{it} a_{jt} SP_{jt} \right), \quad \forall t \in T. \quad (14)$$

We consider two objective functions in the model, one of which minimizes the total cost, and the other minimizes storage quantity. Total cost includes purchasing, component and production, storage, and product cost; they are presented in detail in Eq. (1). The monomial in Eq. (1) is in the order of fixed order price, per order cost, stockholding/storage cost of components, stockholding/storage cost of products, and product cost. Eq. (13) represents the stockholding/storage cost of products, and Eq. (14) represents the stockholding/storage cost of components. In Eq. (13), the first expression in $\max(\cdot)$ represents the production cost, and the second expression is the shortage cost. When the production quantity is non-negative (the shortfall quantity is equal to 0), there is no shortage, and Eq. (13) can reflect either the inventory cost or the shortage cost. Since the production quantity is negative, it indicates a market shortage. The same is true for Eq. (14) for inventory/shortage costs of components. Eq. (2) represents minimizing total storage quantity.

Constraint (3) is the component inventory level balance formula. The starting component inventory for a given period is the sum of the starting component inventory for the prior period plus the number of purchased components less the number of components needed to make the product. Constraint (4) indicates the inventory balance constraint at the stockholding level of products. Similarly, the initial stock of a product in a particular cycle equals the initial inventory of a product in the preceding cycle plus the number of purchased components minus the number of components required to produce products and the number of products required by the market. Meanwhile, constraint (5) is sourcing quantity on-demand at the market; the quantity purchased at the time of interruption equals the market demand. Constraint (6) represents that production quantity depends on the manufacturer's production capacity and the manufacturing load rate. Constraint (7) states that the purchased quantity cannot exceed the manufacturer's capacity limit. Constraint (8) means that the quantity purchased plus the opening inventory cannot exceed the warehouse capacity of the component. And similarly, constraint (9) shows that the product produced plus the beginning inventory cannot exceed the product's storage capacity. Constraint (10) enforces the storage constraint in every market. The shortage depends on whether the product inventory exceeds 0. If the product inventory is less than 0, it indicates a shortage of the number of items the market requires. On the contrary, if the product inventory exceeds 0, the shortage quantity is 0. Moreover, constraint (11) guarantees the sourcing quantity of supply i is not less than zero and takes the integer. Constraint (12) defines the value of interrupt probability to be between 0 and 1.

The dual-sourcing model is a complex nonlinear programming problem derived from Eq. (13) and Eq. (14). The inventory/shortage cost is handled following Bertsimas and Thiele [49], and the model is then changed into a bi-objective mixed-integer programming model.

$$\min z_1 = \sum_t \sum_i \lambda_{it} g_{it} + \sum_t \sum_i c_{it} x_{it} + f_t^1 + f_t^2 + \sum_t PM_{jt} e_t \quad (15)$$

objective function (2), subject to:

$$Cb_{t+1} = Cb_1 + \sum_{n=1}^t \left(\sum_i x_{in} p_{in} - kPM_{jn} \right), \quad \forall j, t = 1, 2, \dots, T-1, \quad (16)$$

$$f_t^1 \geq Ph_t \cdot Pb_t, \quad \forall t \in T, \quad (17)$$

$$f_t^1 \geq \sum_j \sum_i p_{it} a_{jt} SP_{jt}, \quad \forall t \in T, \quad (18)$$

$$f_t^2 \geq Ch_t \cdot Cb_t, \quad \forall t \in T, \quad (19)$$

$$f_t^2 \geq \sum_j \sum_i p_{it} a_{jt} SP_{jt}, \quad \forall t \in T, \quad (20)$$

constraint (4), (5), (6), (7), (8), (9), (10), (11), (12).

Among them, Objective (1) becomes Objective (15), and Constraint (16) is obtained by iterative processing of Constraint (3). In addition, Constraint (17) and Constraint (18) are obtained by processing Eq. (13), and Constraint (19) and Constraint (20) are obtained by processing Eq. (14).

3.2. Uncertain scenario model and model solution

The interruption probability is highly uncertain considering the real-world situation, which leads to a typical scenario model complex called the uncertain scenario model. We solve this complex, uncertain scenario model using the interval robust optimization method based on Bertsimas and Sim [50]. This method can obtain a robust solution under uncertain interruption probability and use conservativeness Γ to represent the decision maker's risk attitude, according to Jabbarzadeh et al. [31].

3.2.1. Tackling uncertain interrupt probability

We assume that the interruption probability p_{it} is in $[\bar{p}_{it} - \hat{p}_{it}, \bar{p}_{it} + \hat{p}_{it}]$, in which $\bar{p}_{it}$ reflects the interrupt probability value in the normal scenario and $\hat{p}_{it}$ indicates the deviation of interrupt probability between the uncertain scenario and normal scenario. Hence, the uncertain scenario model is as follows.

$$\min z_1 = \sum_t \sum_i \lambda_{it} g_{it} + \sum_t \sum_i c_{it} x_{it} + \sum_t f_t^1 + \sum_t f_t^2 + \sum_t PM_{jt} e_t \quad (21)$$

objective function (2), subject to:

$$Cb_{t+1} = Cb_1 + \sum_{n=1}^t \left(\sum_i x_{in} \bar{p}_{it} - k PM_{jn} \right) + \sum_j \sum_i \left(q_{it} \Gamma_{it} + \sum_{n=1}^t r_{int} \right), \quad \forall j, t = 1, \dots, T-1, \quad (22)$$

$$Pb_t = Cb_t + \sum_i x_{it} \bar{p}_{it} - \sum_j PM_{jt} - \sum_j D_{jt} + \sum_j \sum_i \left(q_{it} \Gamma_{it} + \sum_{n=1}^t r_{int} \right), \quad \forall t \in T, \quad (23)$$

$$\sum_j \sum_i \left(\bar{p}_{it} a_{jt} SP_{jt} + q_{it} \Gamma_{it} + \sum_{n=1}^t r_{int} \right) \leq f_t^1, \quad \forall t \in T, \quad (24)$$

$$\sum_j \sum_i \left(\bar{p}_{it} a_{jt} SP_{jt} + q_{it} \Gamma_{it} + \sum_{n=1}^t r_{int} \right) \leq f_t^2, \quad \forall t \in T, \quad (25)$$

$$q_{it} + r_{it} \geq \hat{p}_{it} x_{it}, \quad \forall i \in \{1, 2\}, \forall t \in T, \forall n \leq t, \quad (26)$$

$$q_{it}, r_{it} \geq 0, \quad \forall i \in \{1, 2\}, \forall t \in T, \forall n \leq t, \quad (27)$$

constraints (5), (6), (7), (8), (9), (10), (11), (12), (17), (19).

Note that, $p_{it} = \bar{p}_{it} + \hat{p}_{it} \cdot q_{it}$ represents the outage probability with interval range, and $q_{it} \in P = \{ |q_{in}| \leq 1 \mid \forall n \geq 0, \sum_{n=1}^t |q_{in}| \leq \Gamma_{it} \forall t \geq 0 \}$. Eq. (22), Eq. (23), Eq. (24), Eq. (25) are robust counterparts of Eq. (16), Eq. (4), Eq. (18), Eq. (20), respectively. Moreover, q_{it} , r_{int} define the robust model's conservativeness and boundary of uncertain interrupt probability, respectively. This model limits Γ_{it} in $[0, t]$, in which decision-makers can rely on their risk attitude to choose. Then the value of Γ_{it} determines the probability of the corresponding constraint to be infeasible, and it is $1 - \Phi(\frac{\Gamma-1}{\sqrt{t}})$. The element of $\Phi(\cdot)$ is the cumulative distribution function of a standard normal random variable.

3.2.2. Objective function

The solution to dual or multiple objective programming problems is to obtain the most preferred solution, the Pareto-optimal solution, instead of the optimal solution. Generally, multi-objective mathematical programming problem requires the weighting and ε -constraint method [31,51]. The augmented ε -constraint method proposed by Mavrotas [51] provides complete implementation codes and an interactive approach based on the ε -constraint method, which not only improves the model's solvability but also averts the appearance of weak Pareto-optimal.

According to the augmented ε -constraint method, we regard the minimizing cost objective function as the primary function and transform the minimizing storage quantity function into the constraint possessing the upper limit ε . The last transforming model is as follows.

$$\min z_1 = \sum_t \sum_i \lambda_{it} g_{it} + \sum_t \sum_i c_{it} x_{it} + \sum_t f_t^1 + \sum_t f_t^2 + \sum_t PM_{jt} e_t - \text{eps} \times (s/\omega) \quad (28)$$

subject to:

$$\sum_t \sum_j SP_{jt} + s = \varepsilon, \quad (29)$$

constraints (5), (6), (7), (8), (9), (10), (11), (12), (17), (19), (22), (23), (24), (25), (26), (27).

To select the most preferred Pareto-optimal solution among all the solutions under the decision-makers' risk attitude, we use a fuzzy approach following Demir et al. [52] and Esmaili et al. [53]:

$$\alpha_i^l = \begin{cases} 1 & , f_i^l \leq f_i^{\min} \\ \frac{f_i^{\max} - f_i^l}{f_i^{\max} - f_i^{\min}} & , f_i^{\min} \leq f_i^l \leq f_i^{\max} \\ 0 & , f_i^l \geq f_i^{\max} \end{cases} \quad (30)$$

f_i^l shows the value of the i -th objective function in l -th Pareto-optimal solution. α_i^l represents the value of membership function of f_i^l . The overall membership function (in other word, the overall degree of optimality) for l -th Pareto-optimal solution was defined as the total value of balanced purchasing (TVBP):

$$TVBP^l = \sum_{i=1}^n w_i \alpha_i^l, \quad (31)$$

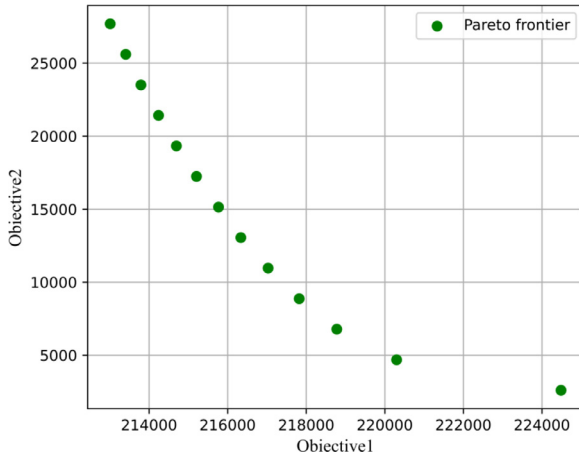


Fig. 2. Pareto frontier in a normal scenario.

where w_i represents the weights of the importance for the i -th objective function. The most preferred solution refers to the Pareto solution with the highest TVBP.

4. Model application and analysis

4.1. Example description

To test the feasibility and effectiveness of the proposed robust model, we take an information transmission equipment manufacturer's production and rough sourcing data in China. Moreover, we combine some studies with setting-related parameters. The enterprise should pass the requirements, procurement, production and delivery process, and procurement stage for each order. According to the production plan, this process determines the procurement and material types levels in a production planning cycle based on the order or demand of a supplier to the source material. However, the company is prone to supply interruption due to changes in the market structure and technological innovation. For this reason, this study designed a dual-source procurement strategy to reduce the cost and shortage loss caused by the interruption.

The enterprise's dual-sourcing problem is to obtain the preferred solution of order assignment under the constraints of capacity, resources, and inventory. There are two available suppliers: the main supply and the backup supply. Related parameters are in the appendix, among which the main supply is relatively cheap and more variable on disruption, whereas the backup supply is more expensive and less variable on disruption. The period of sourcing is 12 months.

4.2. The result of normal scenario

In solving the specific probability of supply disruption on the normal scenario model by the augmented ε -constraint method, we obtain the Pareto frontier point (Fig. 2) and the TVBP corresponding to the different overall membership function values (Table 3).

Fig. 2 shows that the total cost and shortage quantity are inversely proportional. Meanwhile, Table 3 illustrates the TVBP corresponding to the different weights for the total cost objective function. A weight of 0.4 or 0.5 indicates that the TVBP rises and then decreases in the third period, and a weight of 0.6 denotes that the TVBP changes from the sixth period. Nevertheless, the inflection point of the former is greater than the latter, indicating that the degree of optimality of the Pareto-optimal solution is worse, with more concern about the total cost of decision-makers.

We take the weight equal to 0.4, which can obtain the preferred Pareto-optimal solution ($z_1 = 218772$, $z_2 = 6778$) and order allocation

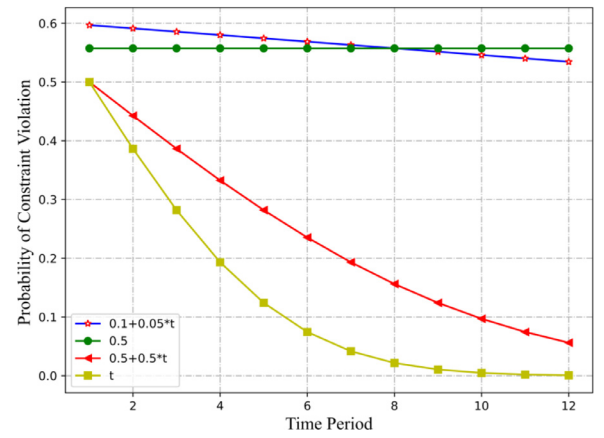


Fig. 3. The probability distribution of violating constraints in four scenarios.

values of the preferred Pareto-optimal solution, as shown in Table 4.

4.3. Analysis of the impacts of suppliers' reliability and decision-maker's risk attitude

Based on the literature review, suppliers' reliability affects the allocation for dual-sourcing. Thus we set the variability of interruption probability to represent the reliability of suppliers: the greater the variability, the more unreliable the supplier. To study the influence of various decision-makers' risk attitudes and variability of interruption probability on two objective functions, we select four risk attitudes and four variabilities for the analysis. The degree of change of interruption probability is calculated as follows.

$$\hat{p}_{it} = \gamma \bar{p}_{it} \quad (32)$$

Note that $\bar{p}_{it}$ is interruption probability on the normal scenario, and γ is the variability of interruption probability. Based on literature [32,54,55], γ equal 5%, 10%, 15%, 20%, separately, and we set Γ_{it} time-related function that reflects decision-makers' attitudes for risk. Thus, the more extensive the value Γ_{it} , the more conservative the risk attitude on decision-makers. The four situations are as follows.

Scenario I. $\Gamma_{it} = 0.1 + 0.05t$,

Scenario II. $\Gamma_{it} = 0.5$,

Scenario III. $\Gamma_{it} = 0.5 + 0.5t$,

Scenario IV. $\Gamma_{it} = t$.

The conservativeness of scenarios II, III, and IV is increasing, and the conservativeness of scenario I is larger than scenario II after the ninth period. Fig. 3 shows the probability distribution of violating constraints of these four scenarios, in which risk attitude is of great conservativeness with increasing conservative interruption probability, thus decreasing the probability of violation of constraints. The probability of violating the constraint in scenario I and II is more significant (0.5), and the probability of violating the constraint in scenario I before the eighth cycle is the largest. In addition, the probability of violating the constraint in scenario III and IV, which are more conservative, tends to decrease, and that of scenario IV, which is the most conservative, decreases to 0 in the last three cycles. Hence, the conservative degree of risk attitude in scenario III and IV will not violate the constraints. The reality is indeed the case. When a decision-maker needs to choose from two suppliers, he or she must consider the following. If the two suppliers are at risk of interruption, the greater is the probability of interruption. Thus, their attitude will be more damaging based on their production conditions, with the most negligible loss to purchase parts from suppliers.

Table 2
Nomenclature; list of indices, parameters, and decision variables.

Indices	
J	Set of multiple demand markets, $j \in J$
T	Set of the period of sourcing, $t \in T$
I	Set of the two suppliers, $i \in I = \{1, 2\}$, 1 refers to the main supplier, 2 refers to the backup supplier
Parameters	
p_{it}	The interrupt probability of supply i in the period of t
D_{jt}	Demand for products at the demand market j in the period of t , which is a random variable subject to a cumulative distribution function of $F(D_{jt})$
c_{it}	Per order price of supply i in the period of t
g_{it}	Fix order price of supply i in the period of t
Q_{it}	Production constraint of supply i in the period of t
Ch_t	stockholding cost of components in manufacturer in the period of t
Ph_t	stockholding cost of products in manufacturer in the period of t
a_{jt}	Storage cost of products at the demand market j in the period of t
V_t	The production capacity of the manufacturer in period t
γ_t	The manufacturer's production load rate in period t
e_t	Product cost in the period of t
k	The proportion for a component to per unit product
PH_t	stockholding capacity of products in manufacturer in the period of t
CH_t	stockholding capacity of components in manufacturer in the period of t
Decision variables	
x_{it}	Purchasing quantity of supply i in the period of t
Cb_t	Inventory level of components at the beginning of the period t
Pb_t	Inventory level of products at the beginning of the period t
SP_{jt}	Storage quantity of product at the demand market j in the period of t
PM_{jt}	Production quantity to product for the demand market j in the period of t
λ_{it}	$\lambda_{it} = \begin{cases} 1, & \text{purchasing products to supply } i \text{ in the period } t \\ 0, & \text{otherwise} \end{cases}$

Table 3
The TVBP corresponding to different overall membership function values.

	z_1	z_2	$w_1 = 0.4$	$w_1 = 0.5$	$w_1 = 0.6$
0	224456	2597	0.613	0.517	0.42
1	220274	4687	0.705	0.652	0.598
2	218772	6778	0.705	0.673	0.641
3	217815	8869	0.688	0.672	0.656
4	217031	10960	0.664	0.663	0.663
5	216328	13050	0.638	0.651	0.665
6	215761	15141	0.607	0.634	0.66
7	215207	17232	0.576	0.615	0.655
8	214692	19323	0.543	0.595	0.648
9	214237	21413	0.508	0.573	0.638
10	213790	23504	0.474	0.55	0.627
11	213406	25595	0.436	0.525	0.613
12	213008	27684	0.4	0.5	0.6

Table 4
Order allocation values of the preferred Pareto-optimal solution during different t .

	1	2	3	4	5	6	7	8	9	10	11	12
Main supply	0	0	0	1133	0	0	0	0	0	0	0	2167
Backup supply	220	0	785	0	709	0	0	0	0	0	0	1007

To compare the robust method and certain probability method under the uncertainty of decision-maker's risk attitude and the influence of suppliers' reliability on differences, we calculate the main objective's difference between the robust method and the particular method for the four risk attitudes with variable degree of the four main interruption probability scenarios. These are robust objective functions relative to the nominal scenario of the main difference (Fig. 4).

Fig. 4 shows that the robustness model can better save cost, and cost-saving is increasingly evident as suppliers' reliability is poorer. However, it is not true that the robustness model performs better the more conservative the decision-makers are. As shown in Fig. 4, the cost savings in scenario II, which is less cautious, are more significant than those in scenario IV, which is the most conservative. Furthermore, the robustness model performs better when the supply chain interruption is less predictable. Nevertheless, when supply is stopped, the conservative stance of decision-makers will

result in higher costs. Therefore, when supply is disrupted, it is preferable to purchase from two sources while maintaining a reasonably neutral risk attitude.

To further study the robust performance of the proposed model, we compare the operating efficiency and results of the robust model and certain probability models under different interrupt uncertainty sets. We randomly generated 50 normally distributed samples with the expected value of $\bar{p}_{it}$ from the previous section and a standard deviation of 0.005. According to the study of Sun et al. [26], Lotfi et al. [56], and Kara et al. [57], we compared the number of feasible solutions and the cost target value between the robust model and the deterministic probability model under different supplier reliability and decision-makers' risk attitudes. Table 5 shows the specific results.

In Table 5, the robust model is better than the deterministic model in terms of the number of feasible solutions and the size of the cost target, and our proposed model has good robust performance. Among the 50 samples, the deterministic model can only solve 20 of them, while the robust model can obtain feasible solutions in all different uncertainty sets. In the best or worst case, the robust model is more cost-effective than the deterministic model. For example, the best optimization outcomes can save 7633 units of cost and the poorest can save 927 units of cost when the supplier reliability is 10% and the decision maker's attitude is $0.1+0.05t$. Another observation is that the robust model performs better under higher uncertainty (less reliable suppliers) and a relatively neutral risk attitude.

4.4. Analysis on balance between cost and storage

To analyze the balance between two objective functions, the comparison of the TVBP is used here. If when the variability of interruption probability equals 15%, Γ_{it} is $0.5 + 0.5t$, and the weight for the total cost objective function is equal to 0.4, 0.5, and 0.6, respectively (Figs. 5, 6, and 7).

Fig. 5 indicates the TVBP declines in the second period, regardless of the weight: the lesser the weight, the more prominent the trend of decline. In other words, less attention paid to the cost causes a more obvious decreasing trend of the optimal Pareto solu-

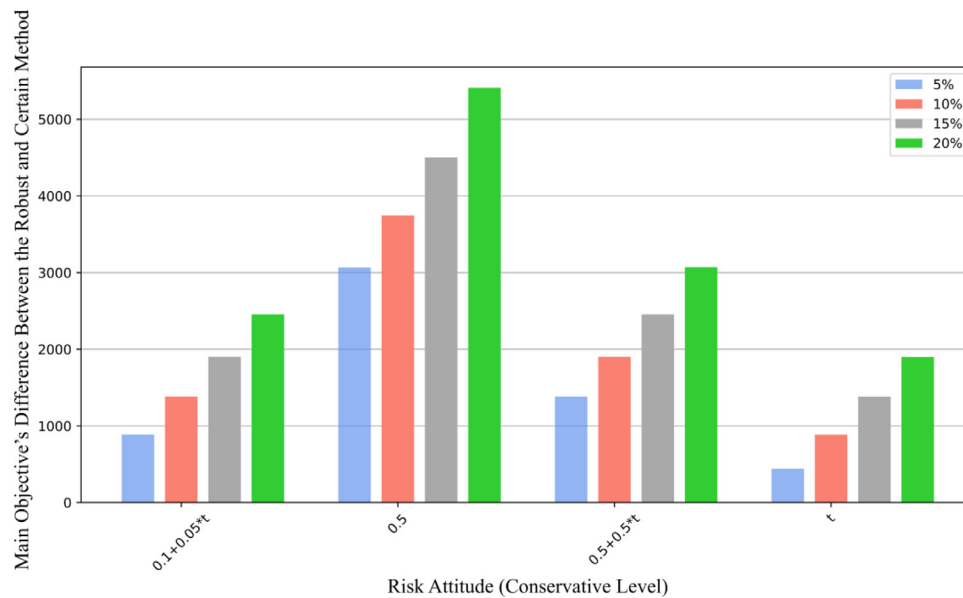


Fig. 4. Difference value of robust primary objective in different risk attitudes and variable case.

Table 5

Robust model is compared with deterministic probability model.

Suppliers' reliability	Decision-maker's risk attitude	Feasible solutions number	Average of main objective values	Standard deviation	Best optimization results	Worst optimization result
5%	$\Gamma_{it} = 0.1 + 0.05t$	50	217929.38	64.05	6611	866
	$\Gamma_{it} = 0.5$	50	217039.68	51.40	6609	966
	$\Gamma_{it} = 0.5 + 0.5t$	50	217931.34	52.79	6613	847
	$\Gamma_{it} = t$	50	217931.76	53.61	6635	894
10%	$\Gamma_{it} = 0.1 + 0.05t$	50	216930.46	49.17	7633	927
	$\Gamma_{it} = 0.5$	50	216050.4	54.77	7599	1097
	$\Gamma_{it} = 0.5 + 0.5t$	50	216933.88	55.75	7633	963
	$\Gamma_{it} = t$	50	216950.24	50.50	7618	962
15%	$\Gamma_{it} = 0.1 + 0.05t$	50	215945.78	52.44	8384	1028
	$\Gamma_{it} = 0.5$	50	215042.68	40.64	8423	1167
	$\Gamma_{it} = 0.5 + 0.5t$	50	215920.2	54.37	8651	1049
	$\Gamma_{it} = t$	50	215942.38	55.37	8603	990
20%	$\Gamma_{it} = 0.1 + 0.05t$	50	215436.32	59.38	8630	1007
	$\Gamma_{it} = 0.5$	50	215010.94	43.05	9329	1203
	$\Gamma_{it} = 0.5 + 0.5t$	50	215735.92	45.37	8626	986
	$\Gamma_{it} = t$	50	215611	48.14	8615	987
Certain probability model		20	224456.4	50.87	-	-

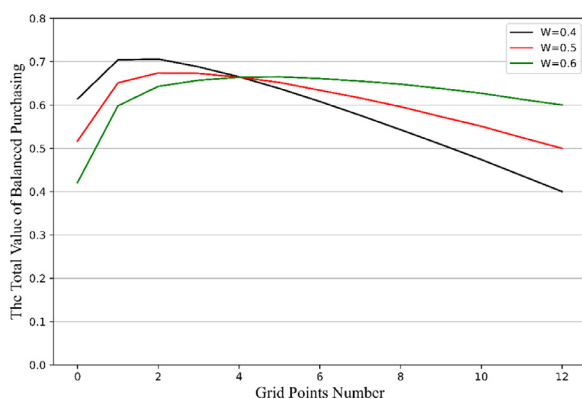


Fig. 5. TVBP corresponding to different weights.

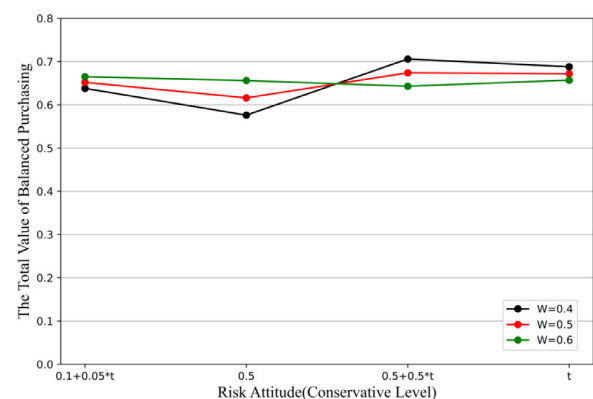


Fig. 6. TVBP corresponding to different decision-makers' attitudes on risk.

tion obtained by the dual-objective function. Moreover, minimizing costs as the main goal is a wise choice.

Fig. 6 shows the influence of different risk conservative degrees on the TVBP. The TVBP is between 55% and 70%, but the conserva-

tive situation of the four risk attitudes has a greater impact on the TVBP with little difference. However, when w equals 0.6, the value of TVBP is more stable in the four risk attitudes. That is, decision-makers with different risk attitudes may adopt the same trade-off when making cost minimization as the primary goal.

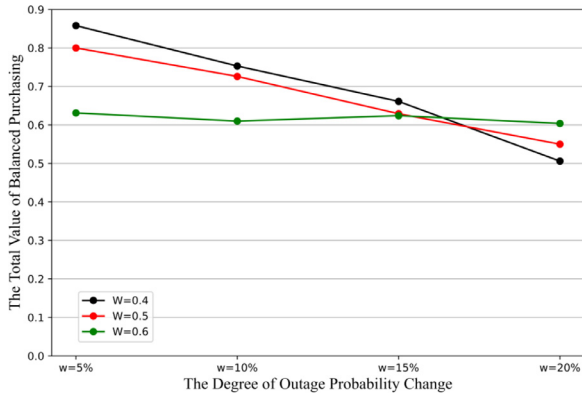


Fig. 7. TVBP corresponding to different variability of interruption probability.

Fig. 7 shows that interruption risk variability has a great influence on TVBP. When the interruption risk variability is 5%, the optimality of the Pareto-optimal solution is the best. Moreover, if the weight is 0.4, TVBP can be close to 0.9. Therefore, when the variability in interruption risk increases gradually, the corresponding TVBP will keep decreasing. As supplier reliability becomes much worse, the optimality of the Pareto-optimal solution is also much worse. Additionally, when the interruption variability is small, the greatest TVBP is in $w_1 = 0.4$, and when the interrupt variability is large, the best weight is 0.6. Thus, When two suppliers are relatively reliable, reducing the cost of the supply chain must be the primary purpose of gaining the supply chain cost advantage.

4.5. Analysis of the strengths of dual-sourcing strategy

In this part, an analysis of the advantages of dual-source procurement is conducted to study. Thus, a comparative analysis of the one-sourcing and dual-source procurement strategies is conducted here. The degree of risk conservativeness of the decision-maker selected here is $\Gamma_{it} = 0.1 + 0.05t$. The model in the previous chapter can solve the objective function value of the dual-source procurement strategy, and the model that one-source procurement strategy can be solved by a robust model subject to the following constraints:

$$x_{2t} = 0 \quad \forall t \in T. \quad (33)$$

Fig. 8 presents the Pareto frontier point data chart corresponding to each interruption uncertain degree of the dual-sourcing method and the non-use dual-source procurement strategy.

In Fig. 8, the dual-sourcing strategy (as opposed to the single-source strategy) significantly reduces cost and market storage in the supply chain under the condition of interruption probability of various variabilities. Additionally, saved cost value is not affected by variability and tends to continuously go up, showing that dual-sourcing has a higher risk tolerance level to save cost and performs well in terms of meeting customer demand. Cost reduction is more significant than the reduction in market scarcity. In other words, the dual-sourcing strategy takes reducing expenditure as the primary goal and solves the problems of product shortage downstream of the supply chain as much as possible in case of supply interruption, which proves that dual-source is a desirable method from an economic point of view.

5. Discussion

In this paper, a dual-objective robust optimization model for dual sourcing strategy on supply interruption risk is established

to prove the stability advantage of the robust method and the economic advantage of dual-source procurement. First, we find that the more prominent unreliability supplier and risk-neutral decision-maker can save more money by adopting the results of the robust technique under different interrupt variability and policymakers' conservative cost estimates. Second, the global optimal values of the Pareto-optimal solution under different interruption variability, cost, and shortage weight conditions are analyzed. Therefore, for reliable suppliers, decision-makers may attach more importance to the shortage minimization goal downstream of the supply chain, but in contrast, they attach more importance to the total cost goal. Finally, by comparing the Pareto solutions of dual-source procurement strategy and single-source procurement strategy, it is proved that the dual-source procurement has a pronounced effect on mitigating the interruption risk.

To reflect the general applicability of the method proposed in this paper, we introduce the data value of He et al. [1] into our model and analyze its dual-sourcing advantages and robust performance. Regarding capacity limits, production costs, and inventory capacity, we set them following the paper of Doganis and Sarimveis [58] and Javanmard and Kianehkandi [59]. Hence, we can obtain the comparison value on single-sourcing and dual-sourcing with different variability and the cost value required to save per unit market shortage in Fig. 9. Besides, the TVBP increment concerning different scenarios and different target weights relative to the nominal scenario is in Fig. 10.

Fig. 9 shows dual-source procurement under the interrupt probability of different variability. Compared with monophyletic procurement cost savings, dual-source procurement shows an economic advantage. With the increase in outage probability, the cost-saving is relatively more, but the cost of compensating for the shortage is increasing. It shows that a double-source purchase has more economic advantage but does worse in satisfying the market than a monophyletic purchase under the condition of unreliable suppliers. In a word, in a supply chain with the risk of supply interruption, obtaining the products needed for production from dual suppliers with different production capabilities and reliability is more practical. It is necessary to compensate for the market shortage and maximize the economic loss caused by customer satisfaction.

Fig. 10 shows the value increment of TVBP of different scenarios and target weights relative to the certain interruption probability. The four interruption variabilities of 5%, 10%, 15%, and 20% are numbered 1, 2, 3, and 4, respectively. The four risk attitudes are numbered 1, 2, 3, and 4, respectively. Thus 16 different scenarios are obtained by combination. The cost target weights are 0.4, 0.5, and 0.6, respectively. This figure integrates the variability, interruption probability, and target weight in the previous section to compare the robust and deterministic methods. In Fig. 10, the TVBP and the cost of the target weight increase as the interrupt probability grow. No matter which interruption is variable, the cost reaches the maximum in the second risk attitude. When the unreliability of suppliers is greater and decision-makers' risk attitude is neutral, robust approaches yield better benefits in the supply chain. In addition, when the weight is 0.6, the result of the robust model is always better than that of the deterministic model. Therefore, the robust method in the supply chain with a dual-source procurement mode can better achieve enterprises' supply risk management objectives from an economic perspective.

Although many researchers are concerned about supplier reliability [14,15], supplier outage scene [6,11], and decision-makers' attitude to risk [9,10]. The comprehensive consideration of these directions needs further in-depth research. Therefore, our study creatively uses a robust optimization method, which is different from other dual-sourcing research that fixes the interruption probability to a particular value [1,14,20]. Hence, the proposed robust so-

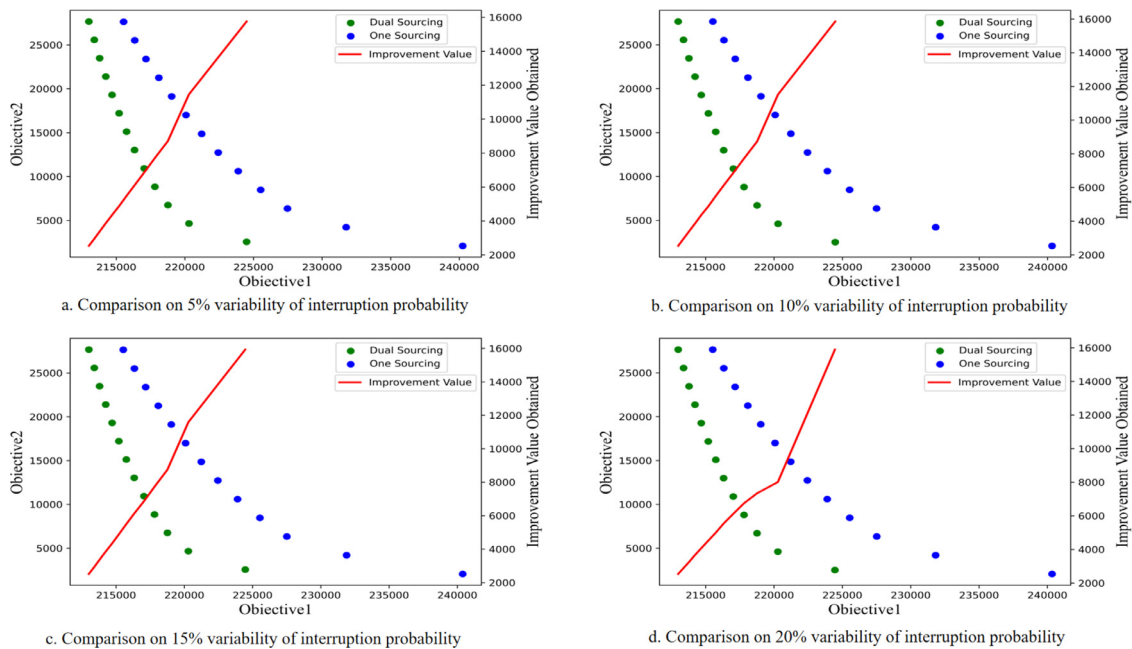


Fig. 8. Comparison of dual-source procurement and single-source procurement strategies under different interruption probability variability.

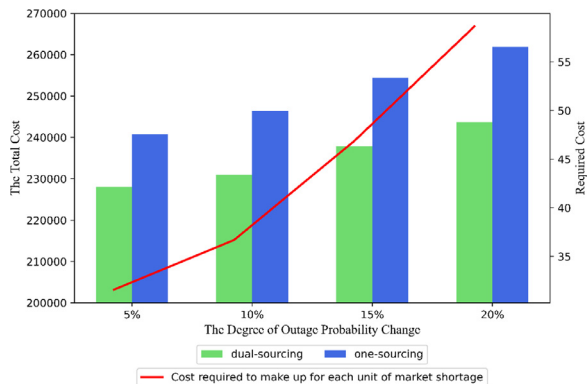


Fig. 9. The comparison value of single-source procurement and dual-source procurement with different variability and the cost value required to save per unit market shortage.

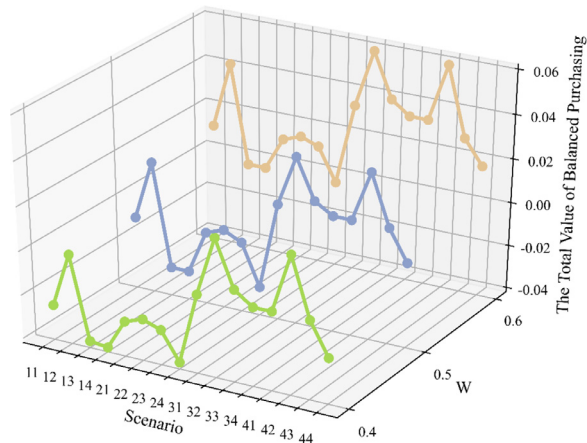


Fig. 10. The TVBP increment with respect to different scenarios and different target weights relative to nominal scenario.

lution is suitable for a certain range of uncertain outage probability and reflects the decision maker's risk attitude. The advantages and application scenarios of the dual-sourcing strategy are explored comprehensively. It provides new thinking and method for supply chain enterprises to make decisions under supply interruption risk.

6. Conclusion and future research

Although supply chain management frequently discusses dual-source procurement, it is not yet well known how an efficient and reliable dual-source subscription program with various decision-makers' risk attitudes may be created in the face of supply disruption concerns. As a result, our paper rise in response to the proper time and conditions. First, we build a dual-sourcing robust optimization model to reduce costs and shortages and use the interval robust optimization method to obtain the robust correspondence mode. Using the augmented ε -constraint method, we rewrite the shortage objective function to obtain a single-objective function in the form of constraints. Additionally, we conducted reliability, risk attitude, dual-target, and dual-sourcing advantage analysis based on a real case. The model performs robustly and efficiently under different uncertainties (unreliable suppliers) and different decision-maker attitudes. Finally, we discuss the broad applicability and feasibility. Some findings are drawn through analysis and discussion, which are summarized as follows: (1) Dual-sourcing strategy based on a robust method takes into account of the equilibrium of cost and market shortage to achieve efficient and successful decision-making, and our method can adapt to various uncertainties and has excellent robustness. (2) Our method is more effective at less reliable suppliers. (3) Though our method can make effective ordering plans under different decision-maker risk attitudes, more efficient and stable results can be achieved under a risk-neutral attitude. The above conclusions also prove that it is meaningful and wise to consider disruption uncertainty and various risk attitudes as dual-sourcing research.

Future research can be extended to the entire supply chain, considering the benefits of suppliers, the game between retailers/manufacturers and suppliers, and other perspectives to explore the impact of supply interruption on procurement strategies. The future research direction of this work can also include extending the proposed model for multiple types of products, more than two sources of supply, and more levels of supply chains. Additionally, the research on the robustness and generalization ability of the model is attractive, promising, and meaningful. Future research can explore the relationship between model robustness and generalization, such as what threshold the robust model reaches will make the model produce poor generalization.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

Declarations of competing interest

None.

CRediT authorship contribution statement

Bing Han: Conceptualization, Methodology, Software, Formal analysis. **Ying Zhang:** Data curation, Writing – original draft, Investigation. **Song Wang:** Visualization, Investigation. **Yongshin Park:** Supervision.

Data Availability

No data was used for the research described in the article.

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Appendix

Table A.1

Table A.1
Example variable data table.

Purchasing cycle	1	2	3	4	5	6	7	8	9	10	11	12
Production demand	D~N (500,5)	D~N (500,5)	D~N (500,5)	D~N (500,5)	D~N (500,5)	D~N (500,5)	D~N (500,5)	D~N (500,5)	D~N (500,5)	D~N (500,5)	D~N (500,5)	D~N (500,5)
production load rate	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%
Order price(Main supply)	0.504	0.52	0.504	0.52	0.504	0.52	0.504	0.52	0.504	0.52	0.504	0.52
Order price(Backup supply)	0.72	0.78	0.72	0.78	0.72	0.78	0.72	0.78	0.72	0.78	0.72	0.78
Order fixed price (Main supply)	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Order fixed price (Backup supply)	4	4	4	4	4	4	4	4	4	4	4	4
Capacity limit (Main supply)	250	300	250	300	250	300	250	300	250	300	250	300
Capacity limit (Backup supply)	350	300	350	300	350	300	350	300	350	300	350	300
Parts inventory holding cost	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021
Product inventory holding cost	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Product shortage cost	2	2	2.5	2	2	2.5	2	2	2.5	2	2	2.5
Production capacity	500	470	400	423	398	373	348	389	298	273	280	250
Cost of production	1	1	1	1	1	1	1	1	1	1	1	1
Production ratio	1	1	1	1	1	1	1	1	1	1	1	1
Parts inventory capacity	220	290	311	212	312	312	212	312	312	312	312	312
Product inventory capacity	220	290	311	212	312	312	212	312	312	312	312	312
Probability of Outage of Nominal Scene (Main supply)	0.002	0.0012	0.0015	0.001	0.0013	0.0015	0.002	0.0012	0.0015	0.002	0.0012	0.0015
Probability of Outage of Nominal Scene (Backup supply)	0.0001	0.0009	0.0009	0.0001	0.0009	0.0009	0.0001	0.0009	0.0009	0.001	0.0009	0.0009

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